

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, TypeScript

Tools/Frameworks: TensorFlow, PyTorch, Docker, FastAPI, AWS (SageMaker), Kubernetes, TensorRT, OpenCV

Competence: Deep Learning, Computational Geometry, Medical Image Analysis, 3D Segmentation, MLOps, Algorithm Optimization, Numerical Stability, Rigorous Validation, Geometric Feature Extraction, Data Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for **metrological applications**, improving model performance and training efficiency.
- Explored novel architectures and **physics-informed hybrid loss functions** to enhance model **accuracy** and **stability** for robust algorithm behavior.
- Collaborated with PhD researchers on a publication for FLOMEKO conference, contributing to **research and exploration of new methods** in a regulated context.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved **vitiligo segmentation accuracy** to 94% and increased processing speed by 5x using advanced **AI algorithms**, deployed with Docker and FastAPI for medical imaging solutions.
- Managed **MLOps pipelines** on **AWS SageMaker**, addressing **YOLO model data drift** and maintaining **Kubernetes infrastructure**, ensuring algorithm **robustness** and scalability.
- Developed real-time tennis foul detection systems with **TensorRT**, reducing compute cost and time by 6x; created POCs using **YOLO**, few-shot learning, and **CNN** for client demos.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed an AI chatbot application using OpenAI API and relevant LLM tools, demonstrating strong Python **software development skills** for AI-based services.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Engineered robust web scrapers and **ETL pipelines** for structured/unstructured data, automating **data ingestion and cleaning workflows** for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble with AlphaFold for **feature extraction**, increasing **model accuracy** by 4% through custom feature engineering in protein function prediction.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed and rigorously **validated deep learning models** for real-time Bitcoin price forecasting, showcasing expertise in time-series analysis and **numerical stability**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first AI-driven self-playing Chitralli Sitar, combining **signal processing** and **music generation LLMs**.
- Successfully delivered the project, featured on Coke Studio, showcasing innovative application of complex algorithms.

Hugging Face

Open-Source

Contributor

- Co-authored **Zero-shot Image Classification technical documentation** and implementation page on the Hugging Face Hub, contributing to open-source **deep learning** resources.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Pursuing advanced studies in Deep Learning, Computer Vision, and Robotics.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Focused on Machine Learning, Data Structures, and Algorithm Design.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python